Fiscal Year:	FY 2017	**Task Last Updated:**	FY 09/25/2017
PI Name:	Fernandez-Pello, Carlos Ph.D.		
Project Title:	Material Ignition and Suppression Test (MIST) in Space Exploration Atmospheres		
Division Name:	Physical Sciences		
Program/Discipline:	COMBUSTION SCIENCE		
Program/Discipline--Element/Subdiscipline:	COMBUSTION SCIENCE--Combustion science		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	94720-1740	**Congressional District:**	9
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2009 Combustion Science NNH09ZTT001N
Start Date:	05/21/2013	**End Date:**	10/31/2020
No. of Post Docs:	1	**No. of PhD Degrees:**	1
No. of PhD Candidates:	1	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	2	**Monitoring Center:**	NASA GRC
Contact Monitor:	Olson, Sandra	**Contact Phone:**	216-433-2859
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Flight Program:			
Flight Assignment:	NOTE: Extended to 10/31/2020 per PI and S. Olson/GRC (Ed., 3/23/16) NOTE: Extended to 10/31/2015 per NSSC information (Ed., 3/24/15)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:	NNX13AL10A		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	NOTE: Continuation of NNX10AE01G, Material Ignition and Suppression in Space Exploration Atmospheres. Central Objectives: The next generation of NASA spacecraft will use lower pressures and increased oxygen concentrations than current designs. Although there is insufficient knowledge about fire behavior of materials in these Space Exploration Atmospheres (SEA), recent research suggests that material ignitability is increased in SEA. These results imply that next generation spacecraft present a greater fire safety hazard than current designs. Although the most effective fire safety strategy is to prevent ignition, if ignition does occur, it is critical to rapidly extinguish the fire. Therefore, an experimental investigation into ease of ignition and fire suppression characteristics of materials under SEA is proposed. Methods/techniques: The proposed experimental apparatus is simple and easily incorporated in the Combustion Integrated Rack on the International Space Station (ISS). It consists of a small-scale cylindrical flow duct that has fuel samples attached to its inside wall. A cylindrical radiant heater is located at the center of the duct. An oxidizer gas flows at down the length of the tube. Electrical igniters are placed downstream of each fuel sample. Time to ignition and the time between introduction of suppression agent and extinction are measured as a function of environmental conditions. The normal gravity and microgravity experimental programs will be supported by a comprehensive model of piloted ignition and extinction of materials in SEA. Microgravity experiments will be used to validate the model, which in turn will be used to predict the fire behavior of different combustible materials in SEA. Perceived Significance: Currently, there exists no testing methodology specifically designed to determine the fire hazards of materials under those conditions. The proposed work will fill that void and provide additional information about the effect of SEA on the flammability and suppression of materials. It will also provide guidance to interpret and extend current NASA testing procedures to SEA.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The project is intended to enhance the ability to predict ignition, suppression, and spread characteristics of materials in their end use configuration. This approach is called performance based design in terrestrial fire protection engineering. Computational support for this approach requires detailed information about material thermal and combustion properties. Our work on MIST will use experimental data to validate computer models attempting to predict fire behavior using material property estimation techniques. These techniques are currently a focus of research in fire protection engineering.
Task Progress:	Most of the ground work has concentrated in complementing the flame spread tests conducted aboard the ISS under the Burning and Suppression of Solids – II (BASS – II) project. For this purpose, tests have been conducted to replicate the BASS-II tests so that the microgravity and the normal gravity test could be compared to obtain information about the differences in the flame spread characteristics in normal and microgravity. An important result of the comparison is the observation that flames can spread at lower oxygen concentrations in microgravity than in normal gravity. This is an important result because it indicates that the limiting oxygen concentration (LOC) below which flame spread cannot longer occur is lower in microgravity than in normal gravity. This result may have significant implications for fire safety in spacecraft. A paper has been submitted to Scientific Reports: S.L. Link, X. Huang, C. Fernandez-Pello, S. Olson, and P. Ferkul. "The effect of Gravity on Flame Spread over PMMA Cylinders."
Bibliography Type:	Description: (Last Updated: 02/15/2026)
Papers from Meeting Proceedings	Link KS, Huang X, Fernandez-Pello C, Olson S, Ferkul P. "The Effect of Gravity on Flame Spread over PMMA Cylinders in Opposed Flow with Variable Oxygen Concentration." 46th International Conference on Environmental Systems, Vienna, Austria, July 10-14, 2016. 46th International Conference on Environmental Systems, Vienna, Austria, July 10-14, 2016. ICES paper ICES-2016-79. , Jul-2016